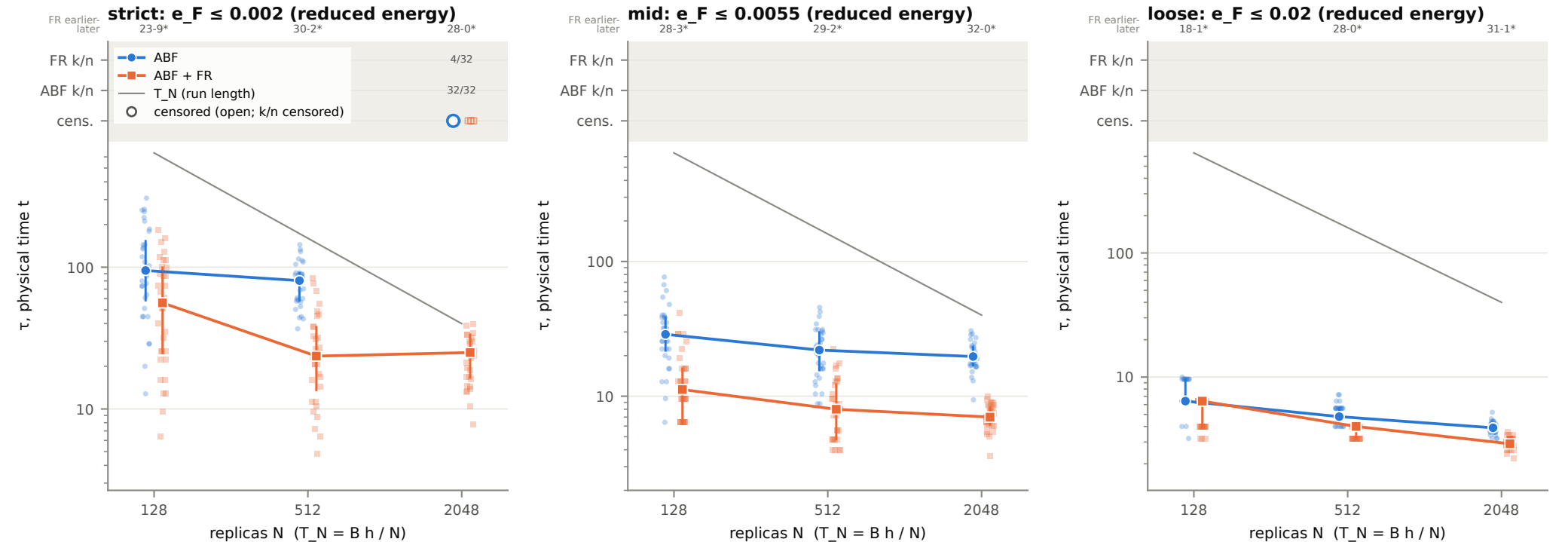


gateway alpha0.5: persistent time to free-energy accuracy, physical time

Equal total budget per arm per seed: $B = N \times n_{\text{steps}} = 3.2768\text{e}+09$ walker-steps at every N ($h = 2.5\text{e}-05$, $T_N = B \cdot h / N$); 32 seeds per (N , method). $\tau(\epsilon)$ = first save after which $e_F \leq \epsilon$ at every later save; never met: censored ($> T_N$, > 1).



gateway alpha0.5 · S4a_tau_free_energy_time · status: all planned (N , seed, method) complete · FR is not defined at $N = 1$ (ABF only, by design).
Markers: seed median (large open marker in the 'cens.' band = median censored); bars: IQR; faint: seeds; censored seeds are open markers in the band, never success at T_N .
Row above each panel: the PAIRED comparison at that N , seeds with FR earlier - FR later than ABF (finite beats censored, both censored = tie; * = exact sign test $p < 0.05$; medians, G and CIs in summary.json / tables.md). Time axis: tau over every save of the run.
Source results/mechanism/matched_free_energy/alpha0.5/analysis/summary.json (generated 2026-10-10T14:19:19Z, eqb_metrics/2, 10000 bootstrap resamples); plot_synthesis/2.